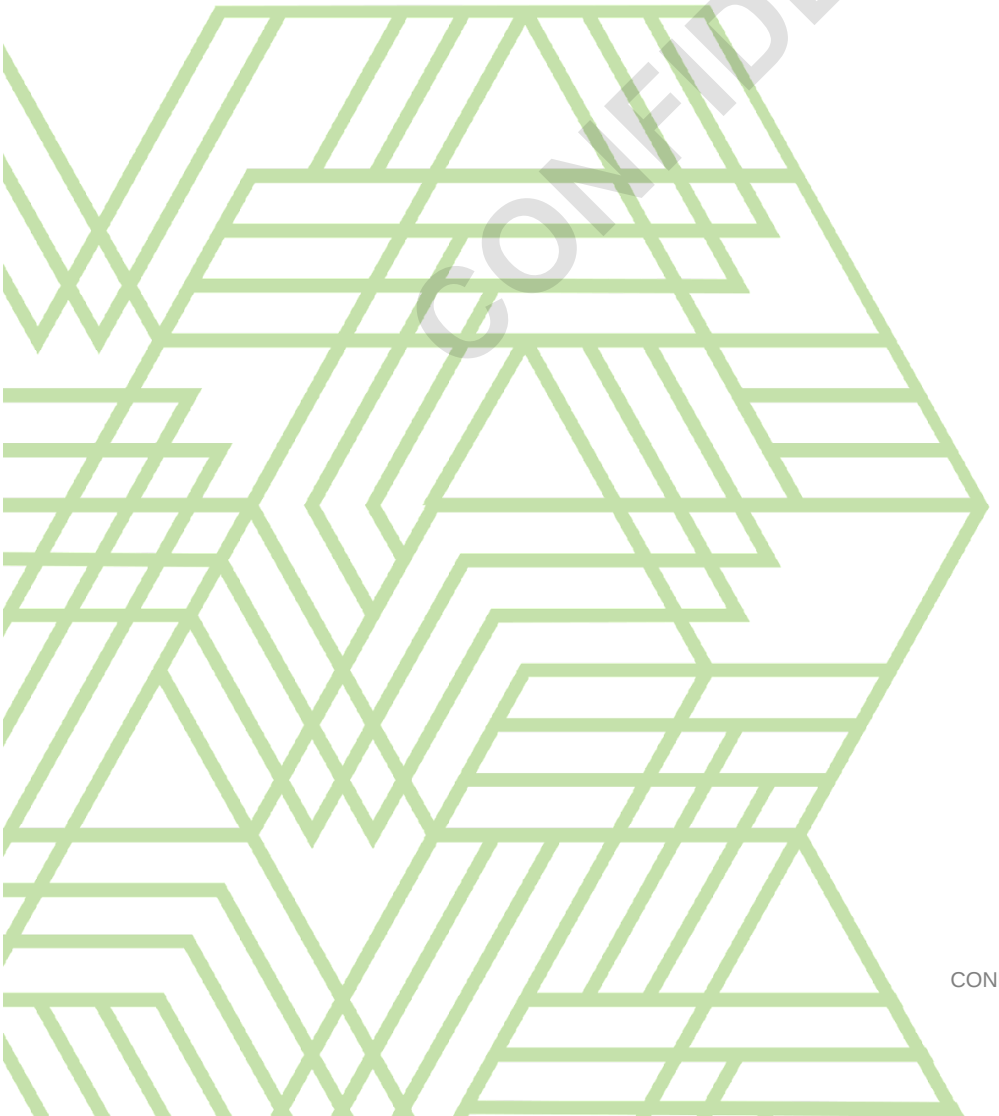
Understanding Low Power Modes for ATM33/e Series

White Paper

SUMMARY: This document describes the functional and power consumption differences among the low power modes and provides an in depth analysis of specific use cases for ATM33/e Wireless SoC Series. This document also describes the possible locations where the user application could reside and how that choice impacts the low power mode selection and operation.

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Table of Contents

1. Overview 3

2. Hibernation and Retention Comparison 4

2.1 Example - BLE_adv 5

3. Hibernation and SoC Off Comparison 6

3.1 Example - BLE_adv 7

Summary 9

Appendix A 10

Revision History 12

List of Figures

Figure 1 - Power Comparison Between Retention and Hibernation
Figure 2 - Hibernation and SoC Off Average Power vs Time Comparison

List of Tables

Table 1 - Power States
Table 2 - Hibernation vs. Retention Comparison
Table 3 - Approximate Retention and Hibernation Characteristics
Table 4 - Hibernation vs. SoC Off Comparison
Table 5 - Approximate Beacon with SoC Off and Hibernation Characteristics
Table A1 - ATM33/e Power Domains and Low Power States
Table A2 - ATM33/e Clock Gating Control and Low Power States

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1. Overview

This document is applicable to the ATM33/e (a.k.a ATM33/ATM33e) series, which consist of:

- ATM33e series
  - ATM3330e 7x7mm package
- ATM33 series
  - ATM3330 7x7 mm package
  - ATM3325 5x5 mm package

The Atmosic ATM33/e Wireless SoC Series support active states and low power states as shown in Table 1 below. Each primary state may contain several secondary states as described in the ATM33e and ATM33 datasheets, as well as Appendix A. Note that in some applications, “hibernate" state may be referred to as “sleep”, while the “SoC Off" state may be called “deep sleep”.

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Table 1 - Power States

Primary Power State	Secondary Power State	PARAM_ID_SLEEEP_ENABLE[1]	Description
Active	Active	0[2]	Connection to Atmosic Interface Board or J-Link chip, APT or DLAll regions of the SoC are powered on. Active power can be optimized by utilizing clock gating registers and/or by putting the Cortex-M33F into Wait for Interrupt (WFI).
	CPU Idle	0[2]	
	Bluetooth LE Deep Steep	1[2]	
	Bluetooth LE Deep Sleep + CPU Idle	1[2]	.
Retention	Retain Drop	2 or 3	Retain RAMs, flip-flops, I/Os states. Wakeup settings must be programmed before transitioning into this state.
Hibernation	Hibernate	4	Powers down system memory. Retains minimal amount of flip-flop state. Retains I/O state. Wakeup settings must be programmed before transitioning into this state.

[1] Refer to the SDK User Reference Manual

[2] This setting is only used for debug

SoC Off	SoC Off	5	All digital domains are powered down, but the PMU remains on in an ultra low-power state with limited functionality.
Powerdown	PWD	Not Applicable	The PWD pin is asserted and all power domains including PMU are completely shut off

In beaconing applications, ATM33/e transmits beacons periodically and remains idle in between those beacons. During the inactivity, ATM33/e can be programmed into one of the three primary low power states: Retention, Hibernation or SoC Off to minimize the power consumption.

In increasing order of steady state power consumption, the ranking is SoC Off then Hibernation then Retention. However, each mode comes with different restart overheads and programing flexibilities which make the best mode dependent on the use case scenarios.

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2. Hibernation and Retention Comparison

In the Retention state, potentially all system memory state and all flip-flop state is retained. Hibernation state is an alternative to Retention state that achieves a lower power than Retention by powering down the system memory and by retaining only a minimal amount of flip-flop state. The time it takes to return from the selected low power state to an active state is different between Retention and Hibernation with Hibernation taking longer. Table 2 below compares the pros and cons between the two states.

Table 2 - Hibernation vs. Retention Comparison

	Hibernation	Retention
Steady State Power Consumption	Pro: Hibernation consumes less power because most digital logic circuits and all memory are disconnected from power.	Con: Retention consumes more power because some digital logic circuits and all configured RAM are retained in a dormant state.
Restart overhead	Con: Waking from hibernation requires initialization that comes with a system reboot.	Pro: Waking from retention requires no initialization since the system continues from its retained state.
Data retention	Con: User data can only be retained in	Pro:

	Hibernation	Retention
	unused sensor hub registers or by overriding control signals for one of the RAM.	All user data in the RAM is automatically retained.
Wake up sources	Wake up sources Both modes offer the same wake up sources in keyboard detection, mouse movement detection, timer, wakeup receiver, and multiple GPIOs.	

2.1 Example - BLE_adv

The ATM33 SDK contains a Bluetooth beacon application with pre-configured parameters (e.g. beacon interval and low power state selection) that are customizable by NVDS tags loaded with the application code. In this use case, the application is programmed to transmit the beacon three times and then goes to one of the low power states for a programmable duration. The cycle repeats in this fashion indefinitely. Table 3 shows the current consumption profile characteristics for both low power modes.

Table 3 - Approximate Retention and Hibernation Characteristics

	Beacon with Retention	Beacon with Hibernation
Average Low-power Current Isleep (µA)	2.1	1.3
Average Active/Beacon Current[3] Iactive (µA)	1300	1450
Total Time of Active/Beacon Tactive (ms)	6.8	15.2
Time from Wakeup to Beacon (ms)	3.2	11.9

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Power as a function of increasing beacon interval for both cases was computed using the equations:

$$E_{active} = I_{active} \cdot V_{batt} \cdot T_{active}$$

$$P(t) = \frac{E_{active} + I_{sleep} \cdot V_{batt} \cdot t}{T_{active} + t}$$

[3] All measurements are conducted at 0 dBm transmit power

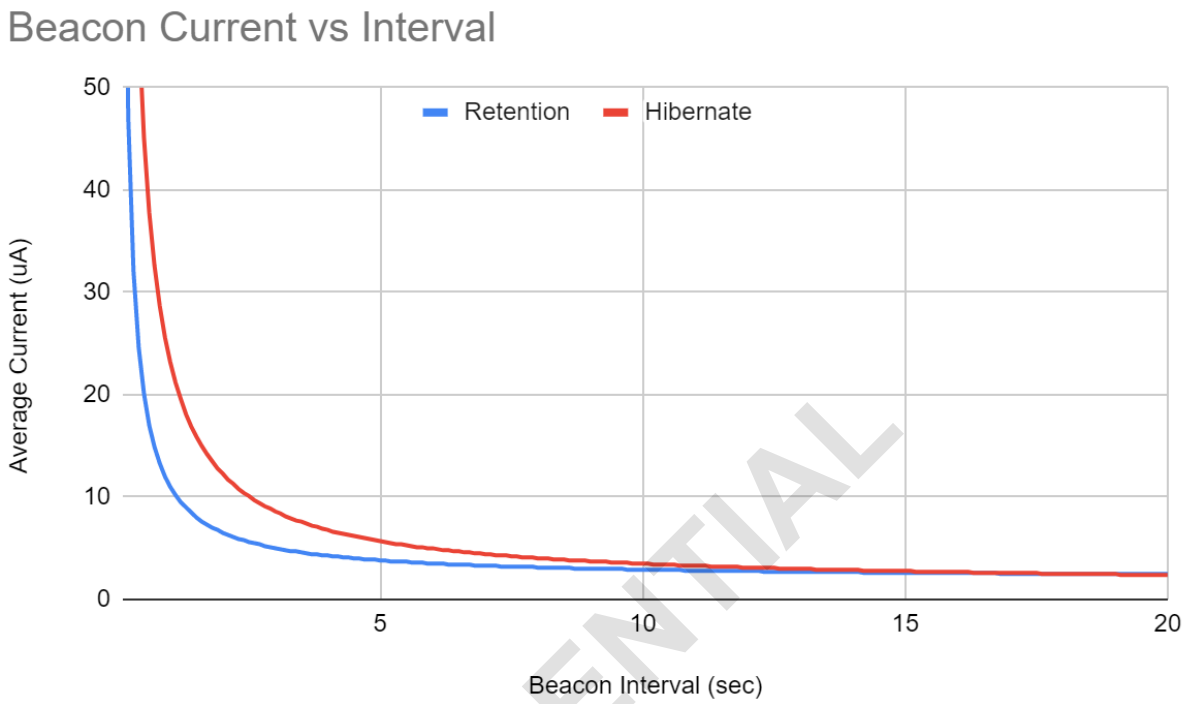


Figure 1 - Power Comparison Between Retention and Hibernation

Based on Figure 1, Retention is more power efficient for duration of < 18 s and hibernation becomes more efficient beyond that.

3. Hibernation and SoC Off Comparison

All state elements are reset upon coming out of either hibernation and SoC Off modes, therefore the SoC system reboots upon coming out of either mode. Table 4 shows the pros and cons between the two modes:

Table 4 - Hibernation vs. SoC Off Comparison

	Hibernation	SoC Off
Steady State Power Consumption .	Con: Hibernation consumes more power because PMU supplies are not shut off and some digital logic circuits are still active.	Pro: SoC Off consumes less power because some PMU supplies are shut off and only certain PMU logic circuits are still active.
Restart Overhead	Pro: Waking from hibernation only requires initialization that comes with a system	Con: Waking from SoC Off requires initialization that comes with a system

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	reboot.	reboot as well as significant inrush current that comes with restoring PMU supplies.
Data Retention	Pro: User data can be retained in unused sensor hub registers or by overriding control signals for one of the RAM.	Con: No user data can be retained.
Wake up sources	Pro: Wake up sources include keyboard detection, mouse movement detection, timer, wakeup receiver, and multiple GPIOs.	Con: SoC Off has a smaller subset of wakeup sources which include a timer, a single GPIO trigger, and a low power analog comparator.

Table 5 shows the current consumption profile characteristics for both low power modes.

3.1 Example - BLE_adv

Similar to the comparison with Hibernation and Retention, this section uses the same BLE_adv example but does a comparison between hibernate with SoC Off states.

Table 5 - Approximate Beacon with SoC Off and Hibernation Characteristics

	Beacon with SoC Off	Beacon with Hibernation
Average Low-power Current Isleep (µA)	0.60	1.3
Average Active/Beacon Current Iactive (µA)	14.6	1450
Total Time of Active/Beacon Tactive (ms)	14.0:	15.2
Time from Wakeup to Beacon (ms)	11.2	11.9

Power as a function of increasing beacon interval (t) for both cases was computed using the equations:

$$E_{active} = I_{active} \cdot V_{batt} \cdot T_{active}$$

$$P(t) = \frac{E_{active} + I_{sleep} \cdot V_{batt} \cdot t}{T_{active} + t}$$

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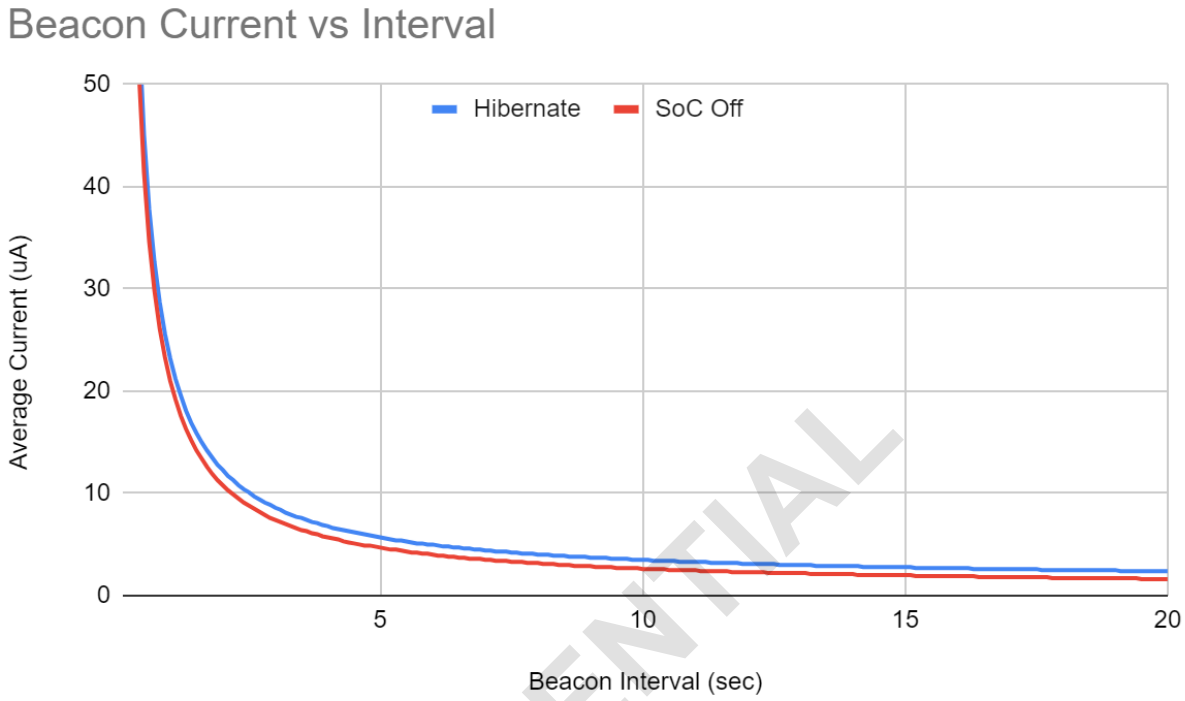


Figure 2 - Hibernation and SoC Off Average Power vs Time Comparison

Based on Figure 2, SoC Off is always better than Hibernate, if power consumption is the only criteria.

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Summary

This white paper describes the trade offs between the three supported sleep states: Retention, Hibernation, and SoC Off. Using a BLE_adv example, we show that one sleep state may be more optimal than the others depending on the interval between the transmitted advertisements.

For ATM33/e, when comparing Hibernation and Retention, Retention is more efficient when the interval of inactivity is < 18 s. Between Hibernation and SoC Off, SoC Off is always more efficient if power consumption is the only criteria used.

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Appendix A

This appendix provides more details on how the power domains and control gating affect each power state.

Table A1 - ATM33/e Power Domains and Low Power States

Primary Power State	Secondary Power State	PARAM_ID_SLEEP_ ENABLE	Power Domains								
			BLE (RF, Link Controller , Modem)	CPU (M33F+P eripheral - RF)	RRAM	System ROM	System RAM 128 KB	Exchange Memory 32 KB	Top 32 kHz	PSEQ	PMU
Active	Active	0	On	On	On	On	On	On	On	On	On
	CPU Idle	0	Off	On	On	On	On	On	On	On	On
	BLE Deep Sleep	1	Off	On	On	On	On	On	On	On	On
	BLE Deep Sleep + CPU Idle	1	Off	On	On	On	On	On	On	On	On
Retention	Retain	2	Registers retained	Registers retained	On	Off	16-128 KB retained	16-32 KB retained	On	On	On
	Retain_Drop	3	Registers retained	Registers retained	On	Off	16-128 KB retained *	16-32 KB retained*	On	On	On
Hibernation	Hibernate	4	Off	Off	Off	Off	Off	Off	On	On	On
SoC Off	SoC Off	5	Off	Off	Off	Off	Off	Off	Off	Off	Partially On
	PWD	X	Off	Off	Off	Off	Off	Off	Off	Off	Off

*VDD lowered

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Table A2 - ATM33/e Clock Gating Control and Low Power States

Primary Power State	Secondary Power State	PARAM_ID_SLEEP_ENABLE	SW Clock Gating Control					
			16 MHz to BLE	16 MHz to CPU	32 KHz	IP Core	WURX	GADC
Active	Active	0	On	On	On	Yes	Yes	Yes
	CPU Idle	0	On	WFI	On	Yes	Yes	Yes
	BLE Deep Sleep	1	Off	On	On	Yes	Yes	Yes
	BLE Deep Sleep + CPU Idle	1	Off	WFI	On	Yes	Yes	Yes
Retention	Retain	2	Off	Off	On	Off	Yes	Off
	Retain_Drop	3	Off	Off	On	Off	Yes	Off
Hibernation	Hibernate	4	Off	Off	On	Off	Yes	Off
SoC Off	SoC Off	5	Off	Off	Off	Off	Off	Off
	PWD	X	Off	Off	Off	Off	Off	Off

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Revision History

Date	Version	Description
August 16, 2022	0.51	Updated Table 3 - Approximate Retention and Hibernation Characteristics, Figure 1 - Power Comparison Between Retention and Hibernation, Figure 2 - Hibernation and SoC Off Average Power vs Time Comparison, Table 5 - Approximate Beacon with SoC Off and Hibernation Characteristics, Table A1 - ATM33/ATM33e Power Domains and Low Power States, Table A2 - ATM33/ATM33e Clock Gating Control and Low Power States.
May 13, 2022	0.50	Initial version created

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